Pump Minimum Flow Determination

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1. PURPOSE & SCOPE

This procedure describes how to determine the lowest flow at which a centrifugal pump can run continuously.

This procedure is applicable to both single and double suction impeller pumps, as well as single- or multi-stage horizontal and vertical pumps.

This procedure applies to Nova Milanese plant and La Spezia plant.

2. DEFINITIONS

AOR: Allowable Operating Region

BEP: Best Efficiency Point

MCSF: Minimum Continuous Stable Flow

MCTF: Minimum Continuous Thermal Flow

NPSHa: Net Positive Suction Head available

NPSHr: Net Positive Suction Head required

SSS: Suction Specific Speed

3. GENERAL PROCEDURE - DESCRIPTION

Two main issues affect the lowest flow at which a pump can operate: vibration level and temperature rise of process fluid. Therefore, it is possible to calculate/estimate:

- a Minimum Continuous Stable Flow (MCSF) as per ANSI/API STANDARD 610 (or other industry standard such as ISO, HI). MCSF is the lowest flow at which a pump can run continuously without exceeding the vibrations limits prescribed in the normative for the Allowable Operating Region (AOR).

- a Minimum Continuous Thermal Flow (MCTF), which is the lowest flow allowed for continuous pump operation without being impaired by the pumped fluid's temperature rise.

Note that this second flow can be assumed as worst condition when NPSHa = NPSHr. In general, it is necessary to have NPSHa > NPSHr with a proper margin.

Refer to Annex A for an extensive explanation of Minimum Flow

4. METHODOLOGY, ANALYSIS AND INTERPRETATION OF RESULTS

4.1. Determination of MCSF

There are many ways to estimate the pump's MCSF:

- Historical vibration test results specific to the pump type and size.
- Pump hydraulic design:
  - Risk of cavitation: NPSHa, speed at the impeller's eye diameter u_1, suction specific speed or high suction capability due to inducer.
  - Partload recirculation: hydraulic excitation forces, noise, cavitation.
  - Stability of the Q-H curve.
- Mechanical design and construction:
  - Type of pump and service.
  - Power class of the pump, head per stage or impeller's top speed u_2.
- Liquid characteristics and energy density.

The first method is probably the most reliable. However, test data are not always available and, whenever they are, they must be updated according to the current edition of API 610's vibration limits. Hydraulic design data are not always available as well. Therefore, it is possible to use Bradshaw's method to calculate an initial MCSF, subject to experimental verification. The methodology considers three different contributions:

- Pump specific speed and power.

- High speed operation.
- High SSS.

Please refer to **SOP-461** for detailed instructions and **Mod.414** for calculations.

4.2. Temperature Rise at MCSF

Further investigation on the MCSF value is worth conducting, especially for high head applications (multistage pumps), by calculating the temperature rise of the pumped liquid from suction to discharge nozzle. It can be calculated using the following equation:

$$\Delta T_{MCSF} = \frac{g}{c} H_{MCSF} \left(\frac{1}{\eta_{MCSF}/100} - 1 \right) \qquad \text{Eq. 1}$$

Where:

g is the gravity constant [m/s^2].

c is the specific heat capacity [kJ/kgK].

H_{MCSF} is the pump differential head at MCSF [m].

η_{MCSF} is the pump efficiency at MCSF [%].

Good engineering practices suggest that this delta T shall be less than 10 °C, but it can also be limited by process constraints of the project.

Therefore, it should also be established a minimum flow (MCTF) for which the process' fluid temperature increase across the pump is the maximum acceptable. The MCTF can be higher than the MCSF and, in this case, it can limit the pump's Allowable Operating Region. In most cases however, MCTF is far less than MCSF.

5. ACCEPTANCE CRITERIA

The definition of the Minimum Continuous Stable Flow can be done using Bradshaw's method. Then, it is verified by testing the pump and measuring the vibration level at different capacities, ensuring that the limits stated on Table 8 of API 610 Standard 12th edition (or other reference normative considered) are not exceeded.

Additionally, the temperature rise at minimum flow can be calculated through Eq. 1, and a Minimum Continuous Thermal Flow can be established, considering the project's temperature constraints.

The minimum value between the two capacities is addressed as the pump's minimum flow, which corresponds to the lowest boundary capacity of the pump's Allowable Operating Region.

6. REFERENCES

API STANDARD 610, 12th Edition, January 2021

SOP-461 *- Minimum Continuous Stable Flow (MCSF) method and calculation*

Mod.414 *- Minimum Continuous Stable Flow spreadsheet*

E. M. Araza, Pump Minimum Continuous Stable Flow (MCSF) What is MCSF and how is it determined? CENTRIFUGAL PUMPS – Modern Design and Practices

S. Gopalakrishnan, "A New Method for Computing Minimum Flow," Proceedings of the Fifth International Pump Users Symposium; Texas A&M University, May 1988 pp. 41-47

Johann Friedrich Gulich, Centrifugal Pumps, Fourth Edition by, Springer Nature Switzerland AG 2020

7. ANNEX "A": PURPOSE OF A MINIMUM FLOW

There are minimum flow values for continuous operation, for intermittent operation and for permissible temperature rise. The figure below shows the relationships among the various off-design pump phenomena and minimum flow conditions. The head versus rate of flow curve with indicated phenomena is a variation of S. Gopalakrishnan's one from his paper titled, "A New Method for Computing Minimum Flow".

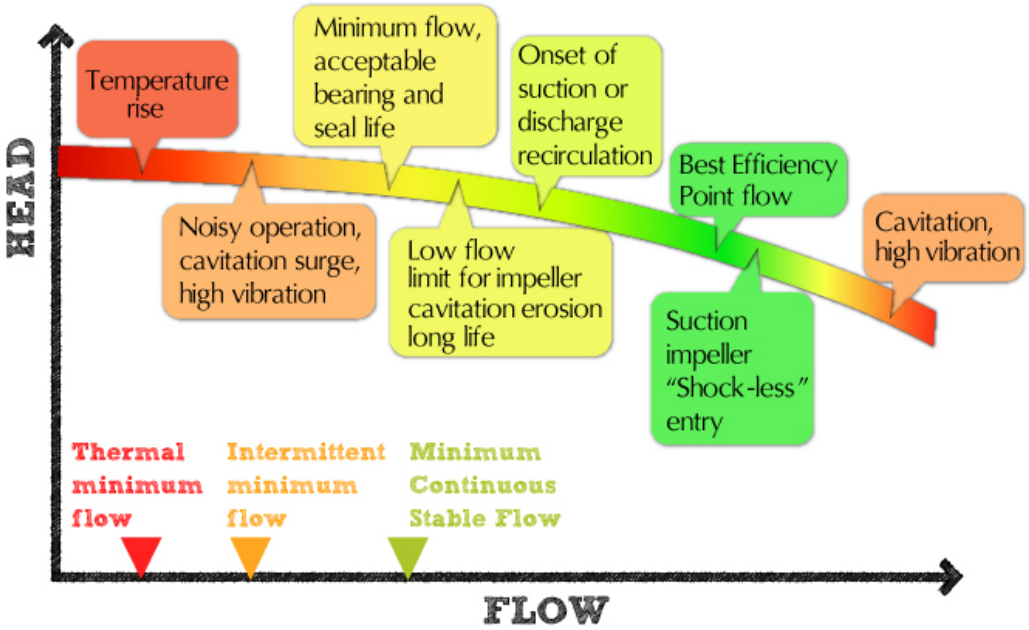

Figure 1: Pump phenomena and minimum flows.

The quoted minimum flow for continuous operation is usually called "Minimum Continuous Stable Flow" or its more common abbreviation "MCSF". It is the flow below which the pump should not be operated continuously. The usual purpose of the MCSF is to achieve satisfactory bearing and seal lives; however, MCSF can be based on other considerations. Any of the following factors must be considered in establishing the MCSF:

- Manufacturer's experience.
- Calculated onset of suction recirculation or discharge recirculation.
- Radial thrust.
- Rise of temperature.

- Cavitation erosion intensity.
- Maximum permissible pressure rise (for system purposes).
- Maximum permissible power rise (high specific speed and axial flow pumps).
- A combination of the above factors or others not listed.

For the hydrocarbon process industry API 610 specification pumps, the value of MCSF is normally coincident with the lower flow limit of the "Acceptable Operating Range" (*Figure 2*) where a specified vibration limit must not be exceeded.

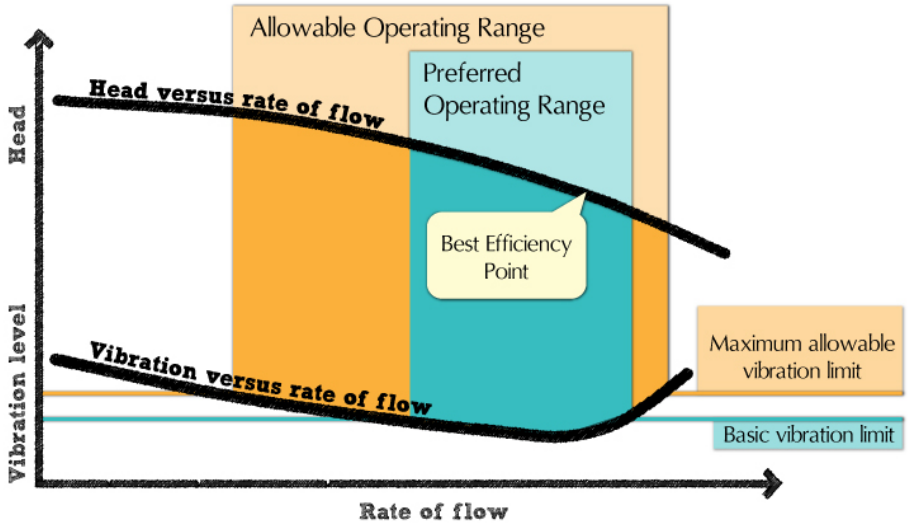

Figure 2: Vibration limits for Allowable Operating Range and Preferred Operating Range.

MCSF is a value that can range roughly from 10% to 80% of the Best Efficiency Point flow depending on pump's size and type, operating speed, impeller suction geometry, liquid density, and other factors. A 2" (50mm) discharge single-stage process pump may have a MCSF as low as 10% of the BEP flow. MCSF is often in the range of 30% to 60% of the BEP flow for process pumps with discharge sizes of 3" (75 mm) and larger. Large mixed flow vertical pumps and very high head-per-stage centrifugal pumps may have a MCSF greater than 60% of the BEP flow. Axial flow pumps have a power curve that rises toward shutoff and the minimum flow may be limited by the power rating of its driver.

For certain high energy pumps, the minimum flow is governed by cavitation erosion damage. The minimum continuous flow for an impeller's erosion life of 40,000 hours is the point in which the system's NPSH Available curve intersects the pump's NPSH Required curve, at lower-than-BEP flow.

The intermittent minimum flow, when specified, is usually given as a percentage of the MCSF. For some applications, the governing value may be based on the temperature rise. For large high energy pumps, the value of the intermittent minimum flow could be, for example, "70% of the MCSF, not to exceed 100 hours per year."

For some applications, a thermal minimum flow or "Minimum Continuous Thermal Flow" is specified based on the permissible liquid temperature rise. MCTF is usually, but not necessarily, lower than MCSF. While a pump thermal minimum flow is not always specified, the end user can readily calculate its value based on the input mechanical power heating up the liquid. The limiting temperature rise is based on a safe margin to prevent flashing of the pumped liquid into vapor, potentially causing the pump's seizure.

The thermal minimum flow is not normally a concern at the pump's start-up if the closed discharge valve is set to begin opening right away. If the margin of the system's NPSHA above the pump's NPSHR is minimal, then the temperature rise conditions at the pump's start-up should be checked carefully.

A few pump applications, such as a vertical turbine jockey pump for maintaining pressure in a large fire sprinkler system, can potentially operate continuously at shutoff while the pump's suction recirculation mixes with the water in the sump in which it operates. The sump acts as a heat sink and a minimal water temperature rise is not a problem. This example is a rare exception to an almost invariable stricture on operating the pump continuously at shutoff.

The purpose of minimum flow is generally to prevent undue wear and tear or damage to the pump. In the real environment of a process or utility plant, a pump is operated at just about any condition demanded by the situation at hand. Thus, there are different pump minimum flows for different purposes.